RAJALAKSHMI ENGINEERING COLLEGE

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S2

Started on	Friday, 28 August 2026, 3:28 PM
State	Finished
Completed on	Friday, 28 August 2026, 3:29 PM
Time taken	36 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs(1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

For example:

Input	Result
3 1 2 3 4 5 6	28

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int compareAsc(const void *a, const void *b) {
    return (*(int*)a - *(int*)b);
}

int compareDesc(const void *a, const void *b) {
    return (*(int*)b - *(int*)a);
}

int main() {
    int n;
    scanf("%d", &n);

    int *arrOne = (int*)malloc(n * sizeof(int));
    int *arrTwo = (int*)malloc(n * sizeof(int));

    for (int i = 0; i < n; i++) {
        scanf("%d", &arrOne[i]);
    }
    for (int i = 0; i < n; i++) {
        scanf("%d", &arrTwo[i]);
    }

    // Sort array_One ascending, array_Two descending
    qsort(arrOne, n, sizeof(int), compareAsc);
    qsort(arrTwo, n, sizeof(int), compareDesc);

    long long total = 0;
    for (int i = 0; i < n; i++) {
        total += (long long)arrOne[i] * arrTwo[i];
    }

    printf("%lld\n", total);

    free(arrOne);
    free(arrTwo);
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 2 3 4 5 6	28	28	✔
✔	4 7 5 1 2 1 3 4 1	22	22	✔
✔	5 20 10 30 10 40 8 9 4 3 10	590	590	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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